ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

BASIC PRINCIPLES OF COMPUTER PROGRAMMING AND COMPUTER LITERACY

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 2 Basic Principles of Computer Programming and Computer Literacy paper 1 · practice a.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Accept answers in the student's own words if the meaning is correct. Accept any correct algorithm, flowchart or code that gives the right result.

SECTION A

QUESTION 1

1.1 1.1.1 D ✓ 1.1.2 A ✓ 1.1.3 C ✓ 1.1.4 C ✓ 1.1.5 A ✓ 1.1.6 D ✓ 1.1.7 B ✓ 1.1.8 D ✓ 1.1.9 B ✓ 1.1.10 C ✓ (10)

1.2 1.2.1 False ✓ 1.2.2 True ✓ 1.2.3 True ✓ 1.2.4 True ✓ 1.2.5 False ✓ 1.2.6 True ✓ 1.2.7 True ✓ 1.2.8 True ✓ 1.2.9 False ✓ 1.2.10 True ✓ (10)

1.3 1.3.1 B ✓ 1.3.2 H ✓ 1.3.3 J ✓ 1.3.4 F ✓ 1.3.5 A ✓ 1.3.6 E ✓ 1.3.7 D ✓ 1.3.8 I ✓ 1.3.9 C ✓ 1.3.10 G ✓ (10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2

2.1 Input (any TWO) ✓✓: keyboard, mouse, scanner, microphone, webcam, touch screen. Output (any TWO) ✓✓: monitor, printer, speakers, projector. Storage (any TWO) ✓✓: hard drive, SSD, USB flash drive, memory card, DVD (6)

2.2 Hard drive: stored on the computer itself ✓ – works without Internet / no data cost ✓. Cloud: stored on remote servers on the Internet ✓ – can be accessed from any device / safe if the computer is lost or damaged ✓. (4)

2.3.1 $=B2*C2$ ✓✓ (2)

2.3.2 $=SUM(D2:D5)$ ✓✓ (2)

2.3.3 $170 + 60 + 250 + 120$ ✓ $= R600$ ✓ (2)

2.3.4 $=MAX(C2:C5)$ ✓✓ (2)

2.4 Any FOUR: use strong passwords and keep them secret ✓; do not open links or attachments from strangers ✓; do not share personal information (ID number, address) ✓; log out on shared computers ✓; use antivirus software; think before posting on social media; report cyberbullying (4)

[22]

QUESTION 3

3.1 Example ✓ per logical step (max 5): 1. Start and move forward. 2. Read the distance sensor. 3. If an obstacle is close, stop, turn left or right and continue. 4. If the far wall is reached (e.g. distance less than 10 cm and no way around), stop. 5. Repeat steps 2–4 until the wall is reached. 6. End. (5)

3.2 Start (oval) ✓; read temperature (parallelogram) ✓; decision "temperature > 30?" (diamond) ✓ with Yes / No labels ✓; Yes → switch fan on (rectangle / output) ✓; No → switch fan off ✓; flow line back to "read temperature" (continuous loop) ✓✓ (8)

3.3 Input: temperature from the sensor ✓; processing: compare the temperature with 30 °C ✓; output: switch the fan on or off ✓ (3)

[16]

QUESTION 4

4.1.1 1 mark per correct row pair: count 1 → total 1; count 2 → total 3; count 3 → total 6; count 4 → total 10; count 5 → total 15 (count ends at 6) ✓✓✓✓✓ (5)

4.1.2 15 ✓ (1)

4.1.3 10 ✓✓ (1 + 2 + 3 + 4) (2)

4.2 Example:

```
INPUT voltage                          ✓
IF voltage < 6.5 THEN                  ✓
   OUTPUT "Charge now"                 ✓
ELSE IF voltage <= 7 THEN              ✓
   OUTPUT "Battery low"                ✓
ELSE
   OUTPUT "OK"                         ✓
ENDIF
```

(6)

4.3.1 It sets pin 13 as an output ✓, then switches the LED on for 0,5 s ✓ and off for 0,5 s, repeating forever (the LED flashes / blinks) ✓. (3)

4.3.2 Once per second ✓ (1)

4.3.3 Change both delays to 250 ✓✓ (delay(250)) (2)

4.4 Any TWO: explain what the code does to other programmers ✓; help you remember your own logic later / make maintenance easier ✓; temporarily disable lines when testing (2)

[22]

QUESTION 5

5.1 Any TWO problems ✓✓: the name does not describe the content or version clearly; hard to find later in a cluttered Downloads folder; easy to open or overwrite the wrong version. Better ✓✓: e.g. Documents › NCV L2 › Robot Project › RobotProject_Report_v2_2026-10-11.docx (descriptive name with version or date, in a project folder) (4)

5.2 Any FOUR: headings and heading styles ✓; table of contents ✓; page numbers / headers and footers ✓; consistent font and spacing ✓; bullets and numbering; captions for pictures; spelling and grammar check; cover page (4)

5.3 Files can be lost through theft, damage, viruses or accidental deletion ✓; back up to a flash drive, external drive or cloud storage (e.g. OneDrive, Google Drive) ✓ (2)

[10]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE